

Optimization

Dario Pescini¹

Università degli Studi di Milano-Bicocca

Dipartimento di Statistica e Metodi Quantitativi

dario.pescini@unimib.it

- By definition, hill climbing halts returning a **local optimum**
- The “quality” of the result is influenced by the chosen **starting solution**
- The algorithm returns a global optimum only if the neighborhood is extended to the whole solution space (unfeasible in most of application cases)

Hill Climbing: Comments

Pros:

- **Ease of use** concerning the implementation and the application
- **Flexibility** in the problem representation and in the choice of the initial conditions

Cons:

- The result is a **local optimum**

To avoid the drawbacks it is possible to:

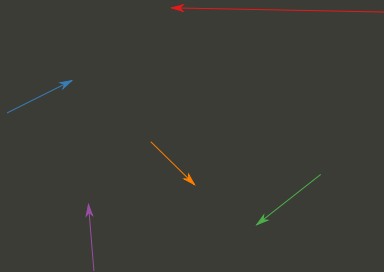
- Repeat the procedure for a huge number of distinct initial conditions
- Define a complex neighborhood structure (if possible)
- Accept (probabilistically) no better fitness solutions (Simulated Annealing)

Particle Swarm Optimizers

- The particle swarm algorithm has been shown to optimize hard mathematical problems in continuous or binary space (Kennedy and Eberhart, 1995; Kennedy and Eberhart, 1997)
- **Particles**, defined as **multidimensional points in parameter space**
- adjust their trajectories **toward** so far best positions
 - of their **own**
 - found by any member **of a topological neighborhood**

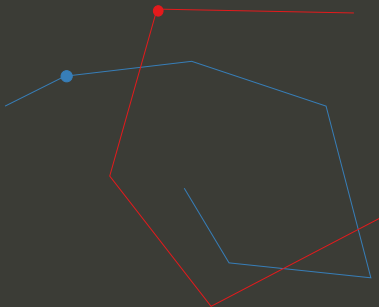
PSO

- A population of particles is initialized with random positions $\vec{x}_{id}$ and "velocities" $\vec{v}_{id}$. At each time-step, each particle is tested in an evaluation function (the fitness)



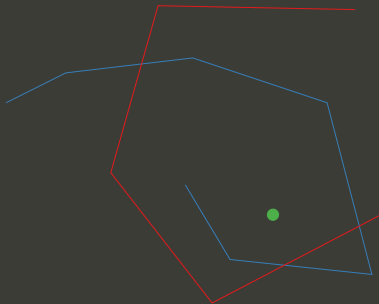
PSO

- A population of particles is initialized with random positions $\vec{x}_{id}$ and "velocities" $\vec{v}_{id}$. At each time-step, each particle is tested in an evaluation function (the fitness)
- The best position found so far by a particle (**individual best**) $\vec{b}_{id}$ is stored.



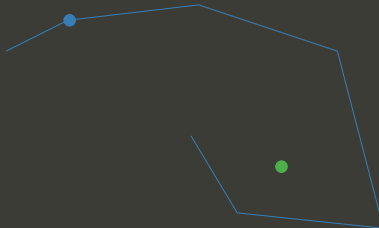
PSO

- A population of particles is initialized with random positions $\vec{x}_{id}$ and "velocities" $\vec{v}_{id}$. At each time-step, each particle is tested in an evaluation function (the fitness)
- The best position found so far by a particle (**individual best**) $\vec{b}_{id}$ is stored.
- As each particle is evaluated, the best position found so far by the swarm (**global best**) $\vec{b}_g$ is stored.



PSO

- A population of particles is initialized with random positions $\vec{x}_{id}$ and "velocities" $\vec{v}_{id}$. At each time-step, each particle is tested in an evaluation function (the fitness)
- The best position found so far by a particle (**individual best**) $\vec{b}_{id}$ is stored.
- As each particle is evaluated, the best position found so far by the swarm (**global best**) $\vec{b}_g$ is stored.



PSO

$$\vec{x}_{id}(t) = \vec{x}_{id}(t-1) + \vec{v}_{id}$$

$$\vec{v}_{id} = w \vec{v}_{id} + c_1 r_1 (\vec{b}_{id} - \vec{x}_{id}) + c_2 r_2 (\vec{b}_g - \vec{x}_{id})$$

- w inertia weight
- c_1 weight of the individual inheritance
- c_2 weight of the social inheritance
- r_1, r_2 random numbers in $[0, 1]$

•

Genetic Algorithms: Darwinian Theory of Evolution (1)

According to Darwin's theory, **natural selection** induces the evolution of the species

In this theory, he points out some fundamental aspects that allow the evolution:

- **Reproduction**

Two individuals can generate an offspring.

- **Environmental Adaptation**

It is the ability of an individual to adapt to the living environment, increasing its chances to survive and to reproduce.

Genetic Algorithms: Darwinian Theory of Evolution (2)

- **Inheritance**
Some features of the parents are passed to the offspring
- **Variation** Some features of the offspring can differ from those of the parents, possibly increasing the adaptation
- **Competition** Due to the limited resources there exists a competition among the individuals to survive

Genetic Algorithms

The starting point is a **population of individuals** (subset of the solution space) and a **fitness function** to assess the quality of each individual.

Iteratively, the “best” individuals are **selected** at each generation, and **variation techniques** are applied to them.

The aim is to get a better adapted set of individuals to the next generation.

Genetic Algorithms

Genetic algorithms accomplish two fundamental steps:

- **Selection:** the process of selecting the **fittest** individuals

Selection mimics the Darwinian principles of survival and competition.

- **Variation:** mainly realized by means of two operators
 - Sexual reproduction (Crossover)
 - Mutation

Variation operators mimic the Darwinian principles of reproduction, inheritance and variation

Selection Operators

Selection operators are not necessarily applied to the whole population. They are applied only to those individuals with higher survival chances.

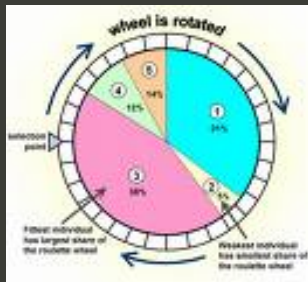
The most widely adopted selection operators are:

- roulette wheel
- ranking
- tournament

Selection Operators: Roulette Wheel

The probability of one individual of being selected is **directly proportional to its fitness**.

$$p_i = \frac{f_i}{\sum_{j=0}^{N-1} f_j}$$



Selection Operators: Ranking

The individuals are sorted according to their fitnesses values.

The probability of one individual of being selected is **directly proportional to its ranking**.



Note:

The most common functions to assign probabilities to individuals are either **linear** or **exponential**.

Selection Operators: Tournament

Features:

- With this operator it is possible **not to evaluate** the fitness values of **each individual** in the population
- Selection takes place randomly picking up k individuals, putting them in a race and choosing the one with the best fitness value
- k is the **tournament size**. It rules the **selection pressure**, weighting how valuable is to have better fitnesses w.r.t. the other individuals.



Selection Operators: Notes

These operators share some common features:

- they allow the individuals with **higher fitness** to be selected with **higher probability** (high fitness corresponds to greater chances to survive)
- each individual within the population has a **probability greater than 0 to be selected** (every individual can survive)
- a **random process** is involved in the selection (better adapted and “lucky” individuals survive)

Variation Operators: Crossover

Crossover: it is the technique that allows to exchange and/or mix features among the population individuals (usually 2). The intent is to generate an offspring that differs from parents but **combines** their features.

The crossover is a **conservation operator** meaning that the new individual mixes already present features only.

Variation Operators: Crossover

The crossover acts in the following way:

- 2 individuals (**parents**) are chosen by means of a **selection technique**
- it chooses the cutting point, **exchange position** among the parents
- the obtained **substrings are combined** in order to obtain 2 new individuals

0110 11001	→	0110 01010
1100 01010		1100 11001

Variation Operators: Mutation

Mutation: it is the technique that allows to modify part of the structure of an individual to generate a new one for the next generation

The mutation is an **innovation operator** meaning that the created offspring might contain features that were not present in the population, thus introducing new genetic material.

Variation Operators: Mutation

The mutation acts in the following way:

- an individual is chosen by means of a **selection technique**
- it chooses the **mutation position**, usually one feature.
- the chosen feature is substituted with another among the allowed ones.

0100**1**1011 → 0100**0**1011

© 2006 The Authors
Journal compilation © 2006 Blackwell Publishing Ltd

- 1 Generate the initial population of N individuals
- 2 Repeat until the halting criterion is satisfied
 - a Compute the individuals fitnesses
 - b Repeat until the new population contains N individuals
 - b.1 Choose a genetic operator
 - b.2 if the operator is crossover select 2 individuals else (mutation) select one individual
 - b.3 Apply the operator
 - b.4 Insert the offspring in the population
 - c Substitute the old population with the new one.
- 3 Return the individual(s) with the best fitness

Genetic Algorithms: Notes

The GA parameters choice are problem dependent. Artcraft work.

- The **structure** of an individual is said to be the **genotype**
- The **fitness** of an individual is said to be **phenotype**

In the GA the **selection** acts on the **phenotype** while the **variation** acts on the **genotype**.

- reproduction
 - heritability
 - variation
- adaptation
- competition

Our PSO Implementation

$$v_{id} = w v_{id} + c_1 r_1 (b_{id} - x_{id}) + c_2 r_2 (b_{gd} - x_{id})$$

PSO1

- **decreasing** w inertia weight plus a Gaussian noise
- c_1, c_2 are independents

PSO2

- **decreasing** w inertia weight plus a Gaussian noise
- c_1, c_2 co-evolve, modified with a Gaussian noise

Our GA Implementation

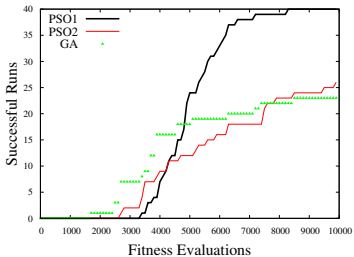
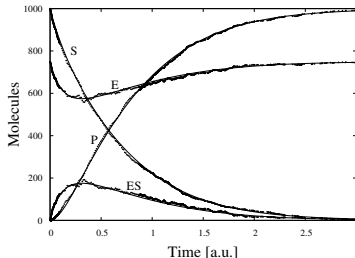
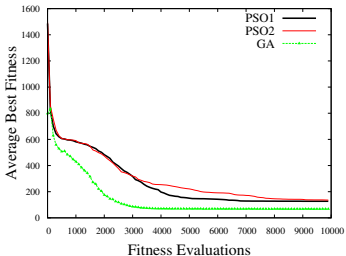
- randomly generate initial population
- $\forall$ generation:
 - $\forall$ individual: evaluate the fitness function
 - apply **elitism**: prevents best fitness individuals to be modified
 - apply **tournament** strategy to select the remaining offspring
 - randomly pick up k individuals and choose the best two
 - apply **crossover** to generate the new individual
 - apply **mutation** according to a small probability value

GA operators { **selection**, **variation** }

Its Operators

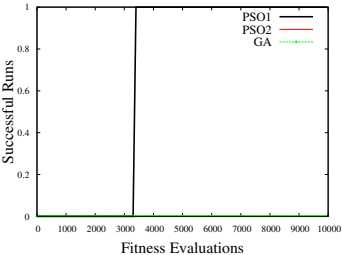
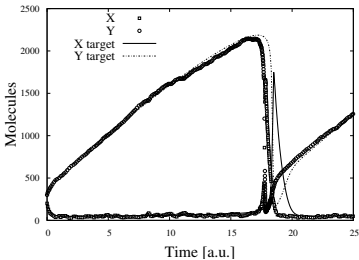
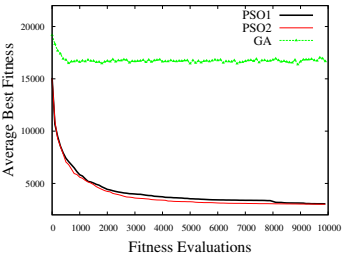
- **One point average crossover**: returns one offspring that contains at each position the average values of the parents chromosomes
- **Elitism**: reduced to one element to preserve the fittest
- **Gaussian mutation**: perturbs an allele with a number drawn from a normal distribution with mean equal to the current value and σ fixed
- **Range mutation**: increments or decrements an allele of a prefixed quantity (that is a fraction of the size of the admissible range)
- **Reinitialization**: changes the value of an allele with an uniformly distributed random number in the admissible range

Comparison GA & PSO: Michaelis Menten



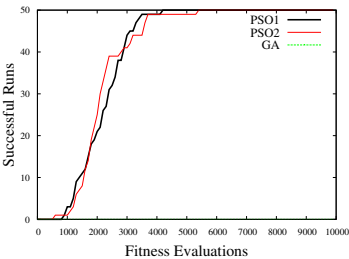
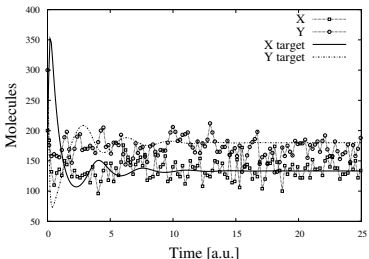
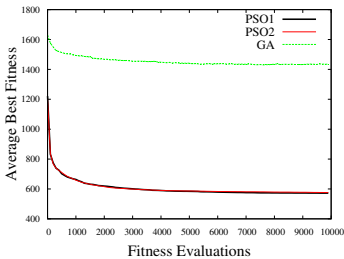
- PS01 best performer
- similar average fitnesses
- dynamics faithfully reconstructed

Comparison GA & PSO: Oscillating Brusselator



- PSO1/2 best performers
- GA worst average fitnesses
- dynamics faithfully reconstructed

Comparison GA & PSO: Dumped Brusselator



- PSO1/2 best performers
- GA worst average fitnesses
- dynamics unfaithfully reconstructed